

Classwork 1

Powers, factorising and algebraic fractions

Grades: [A] [M] [E]

Answers · tutor's working, not an official mark scheme

Show all working. Leave answers with positive indices and fractions in simplest form.

Part A · Powers, roots and factorising

A1 Simplify $\frac{(5x^2)^2}{10x^{-3}}$, leaving your answer with positive indices. [A]

$$\frac{25x^4}{10x^{-3}} = \frac{5x^{4-(-3)}}{2} = \frac{5x^7}{2}$$

A2 Simplify $\sqrt[3]{125p^9}$. [A]

$$\sqrt[3]{125} = 5, \quad 9 \div 3 = 3, \quad \text{so } 5p^3$$

A3 Factorise completely $9x^2y - 6xy^2$. [A]

$$\text{HCF } 3xy: \quad 3xy(3x - 2y)$$

A4 Factorise $2x^2 - x - 15$. [A]

$$\begin{aligned} ac = -30; -6 \text{ and } 5: \quad 2x^2 - 6x + 5x - 15 &= 2x(x - 3) + 5(x - 3) \\ &= (x - 3)(2x + 5) \end{aligned}$$

A5 Simplify $\frac{x^2 - 16}{x^2 + 2x - 8}$. [A]

$$\frac{(x - 4)(x + 4)}{(x + 4)(x - 2)} = \frac{x - 4}{x - 2}$$

Part B · Combine, solve, change the subject

B1 Write $\frac{3}{x+2} - \frac{x-4}{x^2+2x}$ as a single fraction in its simplest form. [M]

$$\frac{3x - (x - 4)}{x(x + 2)} = \frac{2x + 4}{x(x + 2)} = \frac{2(x + 2)}{x(x + 2)} = \frac{2}{x}$$

u: common denominator $x(x + 2)$ and correct numerator; r: correct fraction, simplified fully

B2 Solve $\frac{3}{x} + \frac{1}{x-2} = 2$. [M]

$$\text{Multiply every term by } x(x - 2): \quad 3(x - 2) + x = 2x(x - 2)$$

$$4x - 6 = 2x^2 - 4x, \quad \text{so} \quad 2x^2 - 8x + 6 = 0, \quad x^2 - 4x + 3 = 0$$

$$(x - 1)(x - 3) = 0, \quad x = 1 \text{ or } x = 3 \quad \text{u: quadratic formed; r: both solutions}$$

B3 Make t the subject of $m = \frac{3t-1}{t+2}$. [M]

$$mt + 2m = 3t - 1, \quad mt - 3t = -1 - 2m, \quad t(m-3) = -(1+2m)$$

$$t = \frac{1+2m}{3-m} \quad \left(\text{or } t = \frac{-1-2m}{m-3} \right)$$

Part C · Show that

C x is a whole number. Show that $(x+1)^3 - (x-1)^3$ is always **2 more than a multiple of 6**. [E]

Words you may use

expand · collect like terms · multiple of 6 · whole number · therefore

Sentence starters (optional): *Expanding the first bracket gives ... Subtracting, ... Since x^2 is a whole number,*
...

$$(x+1)^3 = x^3 + 3x^2 + 3x + 1 \quad \text{MP1}$$

$$(x-1)^3 = x^3 - 3x^2 + 3x - 1 \quad \text{MP1}$$

$$(x+1)^3 - (x-1)^3 = 6x^2 + 2 \quad \text{MP2}$$

x^2 is a whole number, so $6x^2$ is a multiple of 6; therefore the result is 2 more MP3

than a multiple of 6, for every whole number x .

MP1 and MP2 together are r; MP3, a conclusion that names why $6x^2$ is a multiple of 6, makes it t.

Extension

Find every whole number x for which $(x+1)^3 - (x-1)^3 = 56$.

$$6x^2 + 2 = 56, \quad x^2 = 9, \quad x = 3 \text{ or } x = -3$$